

Migrating the GÉANT Network...

... to another vendor with WF0 and Ansible

AUTOCON S

THE NETWORK AUTOMATION CONFERENCE

Context: Géant 1/2



Runs a membership association for Europe's National Research & Education Networks (NRENs)



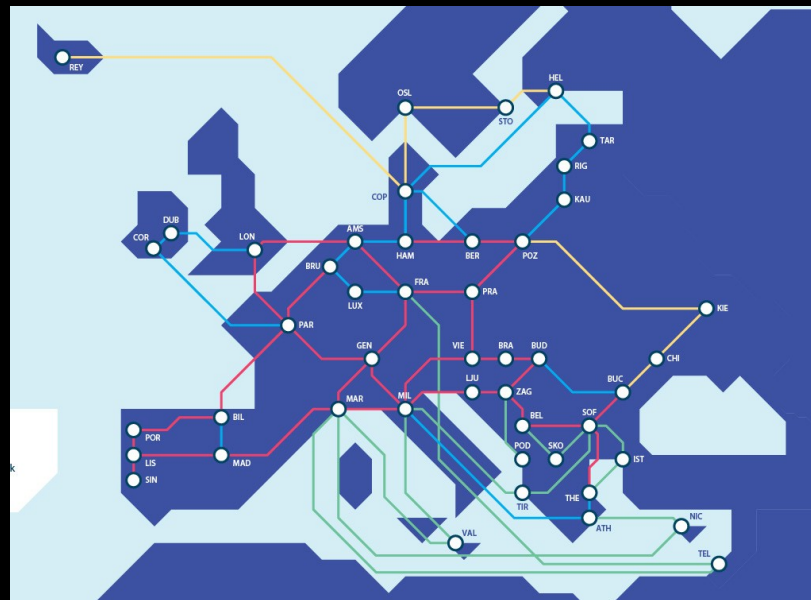
Coordinates and participates in EC-funded projects



Operates a pan-European e-infrastructure

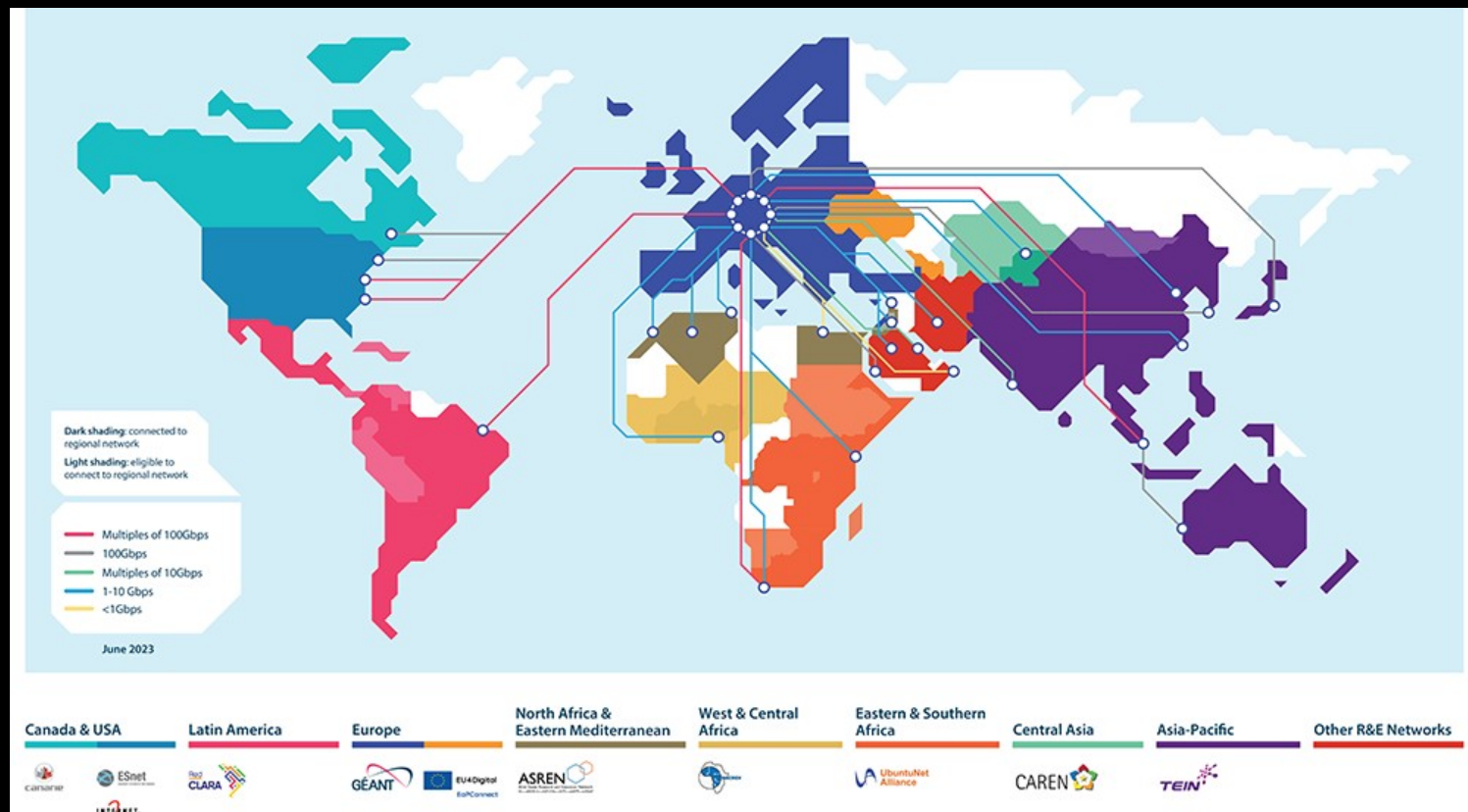


Manages a portfolio of services for research and education



In Europe: 38 NRENs + NORDUnet, supporting 10,000 institutions and 50 million academic users

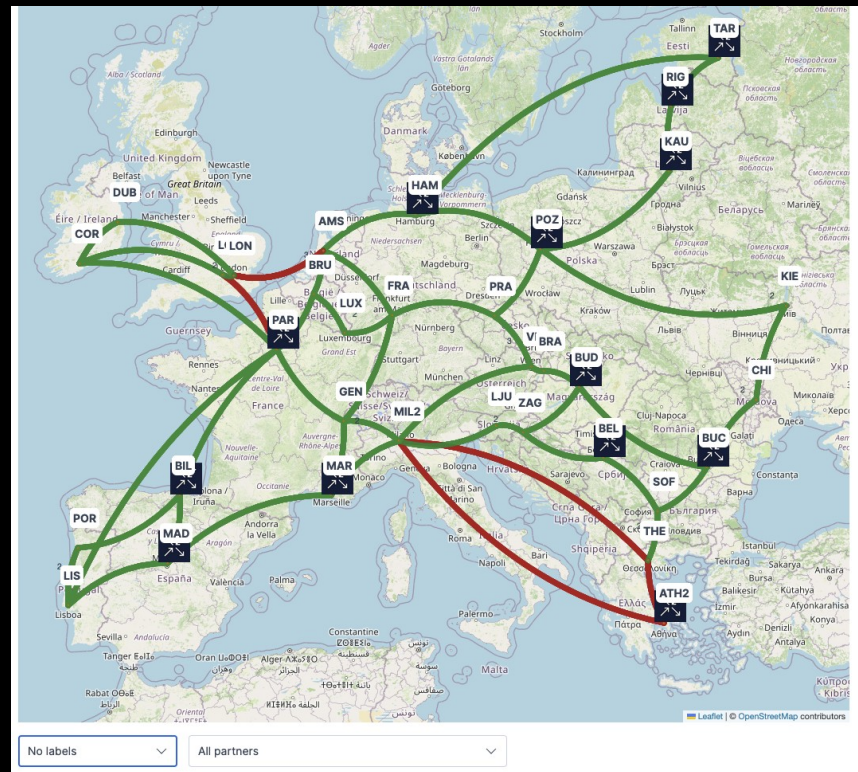
Context: Géant 2/2



Context: the network

Network in numbers (and growing):

- 31 PoPs
- 44 nodes
- >1000 Services:
 - BGP based L3 services
 - Access to different VRF
 - L2 circuits
 - Directly attached Servers
- NB: this diagram comes from GAP ;-)



Migration

April 2023: Start adopting WF0 and building LS0 to pilot Ansible. We start calling it GAP.

June 2023: IP/MPLS tender is awarded

- IP/MPLS layer is re-procured and NOKIA + NOMIOS wins the tender.
- New platform is 7750-SR based on SR-OS (Not SR-Linux)
- 800G everywhere becomes actually possible
- The migration should be completely automated



June 2024: First phase of migration starts

- Nokia routers are deployed initially as P routers (only LSR) via GAP
- First trunk gets migrated to nokia Platform via GAP
- From core to edge

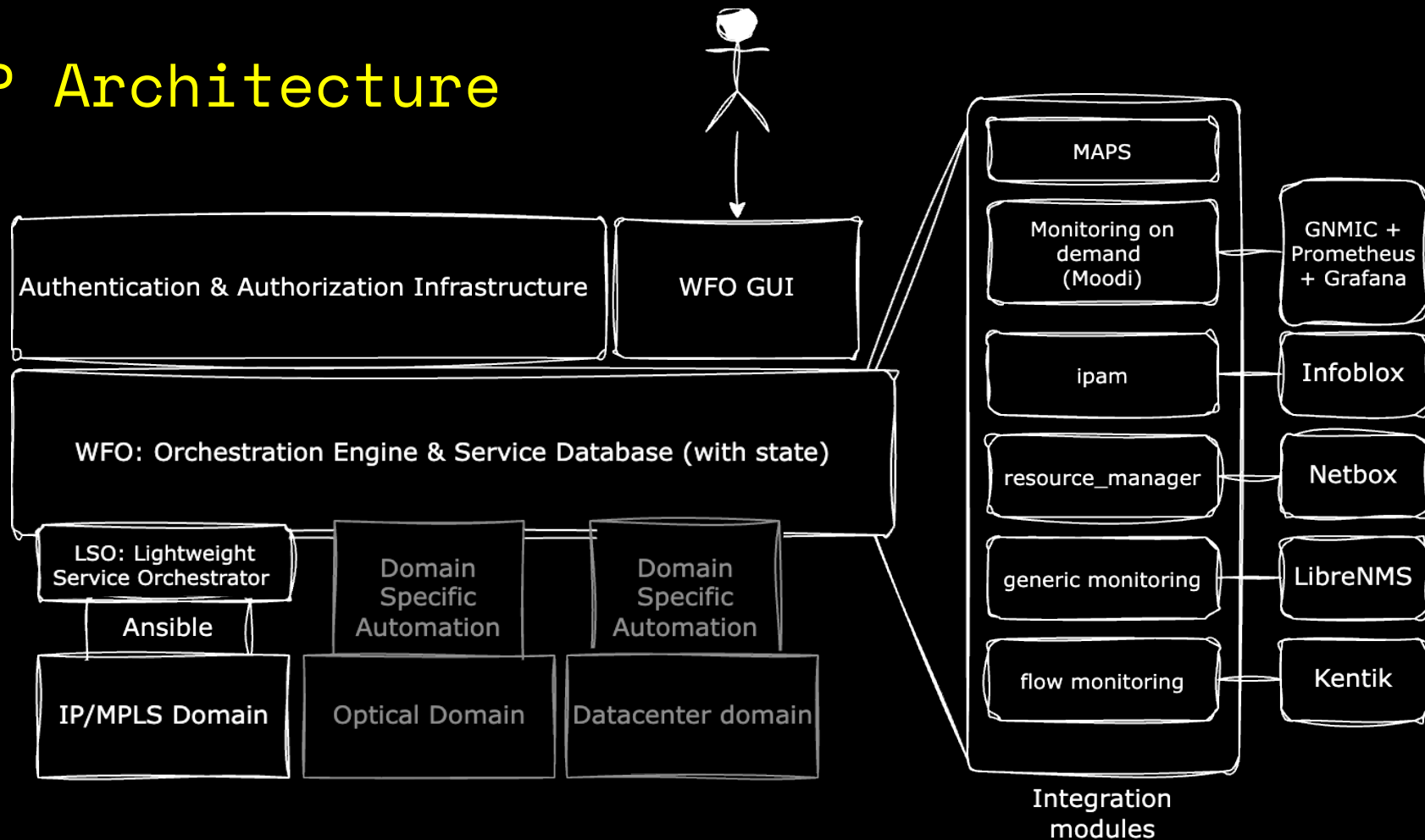


January 2025: second phase of migration starts

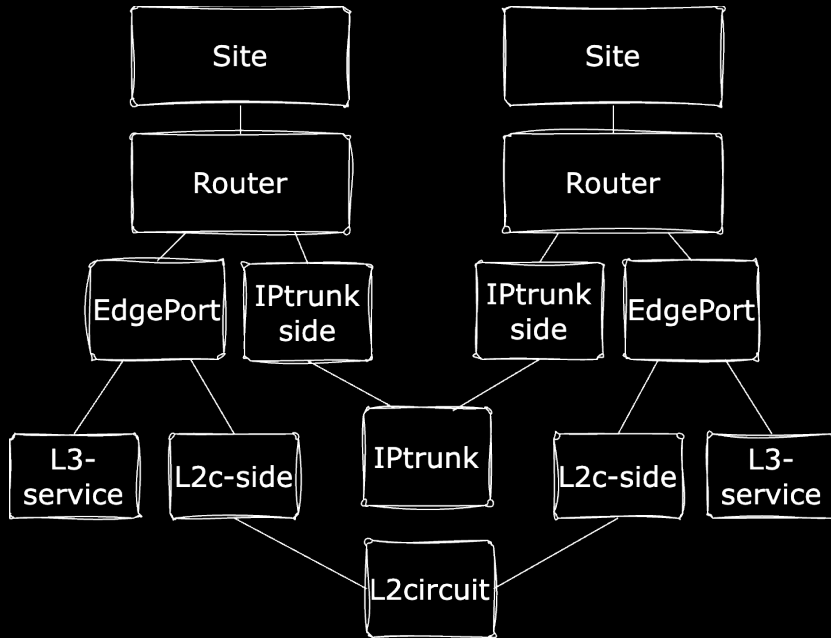
- Services are migrated from Juniper to Nokia. From edge to core
- Initially one service at the time, then one port at the time.

February 2026: Migration is over.

GAP Architecture



Top down approach



- All the config should be there for a reason: it belongs to the node itself or it belongs to a service.

- The whole node configuration is a composition of disjointed parts of configuration that represents node or services.

- Some services involves multiple routers, but we want to focus on the service itself

- One network construct – one Ansible role

- Lifecycle management is embedded

An example: commercial Internet

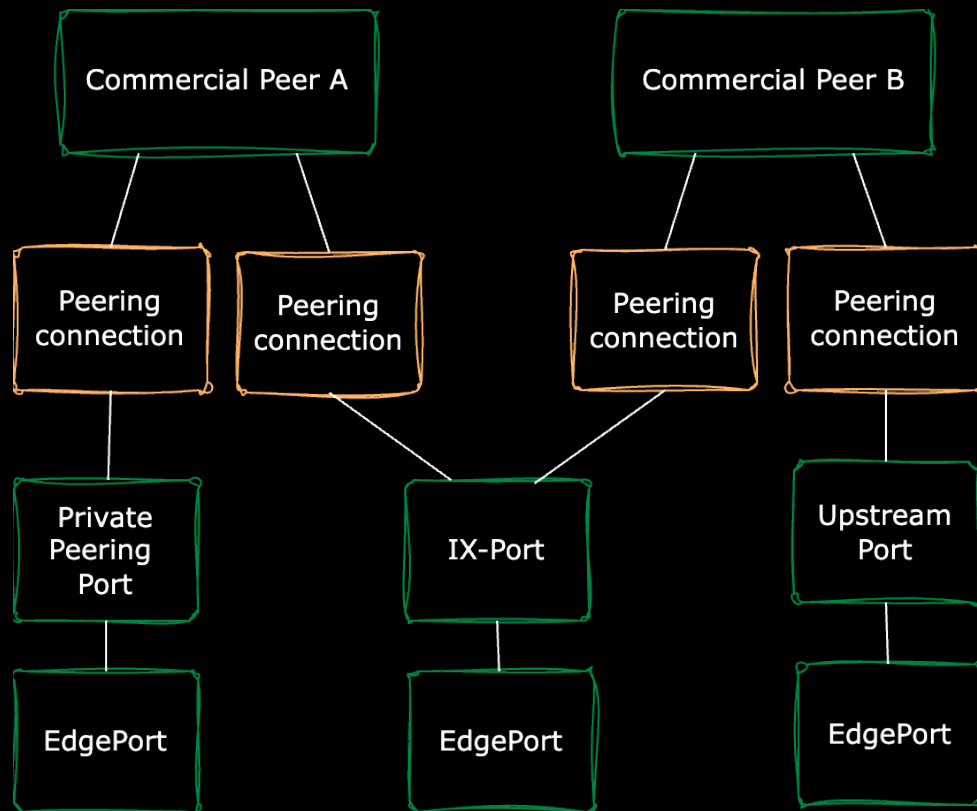
3 possible relations:

- IX peering
- Private Peering
- Upstream Provider

2 possible approaches:

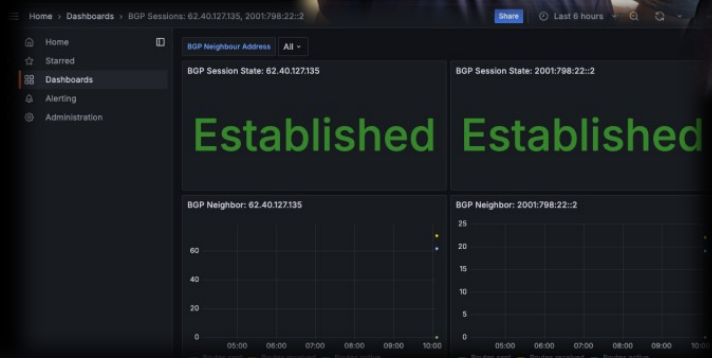
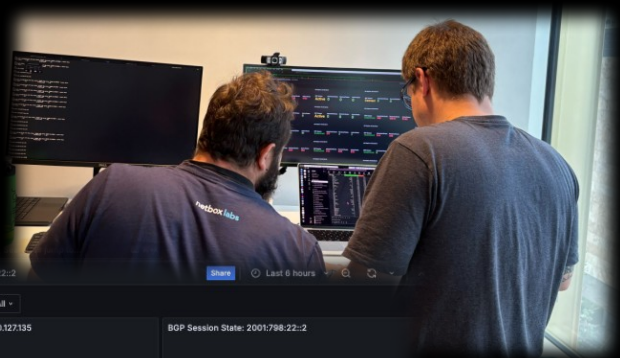
- Port-centric
- Org-centric (chosen)

Peering connections have
lifecycle
(Prov/Active/Disabled)



An example: migrating an IX

- You move the IX port
- GAP finds all the peers
- Moodi starts monitoring them
- Ansible generates the config
- Operator confirms all good
- You can have a beer
- AMS-IX (~120 peers):30 min



Moodi Instances

Process ID	Started	Link
331760bd-f59a-46ba-ac79-a1ffcd77c65f	2025-11-17T09:32:03Z	moodi.gap.geant.org/331760bd-f59a-46ba-ac79-a1ffcd77c65f
451cd196-795c-45f1-82cf-a0ffba04180b	2025-11-05T12:44:25Z	moodi.gap.geant.org/451cd196-795c-45f1-82cf-a0ffba04180b
57e42104-e599-4672-9151-a9b3716d30f1	2025-10-29T12:01:55Z	moodi.gap.geant.org/57e42104-e599-4672-9151-a9b3716d30f1

WFO Programme



Workflow
Orchestrator

We are not doing this alone, and you are welcome to join!

WFO Programme - <https://workfloworchestrator.org/>:

- Public money, public code →
- Mainly R&E, but not only
- 3 level of engagement: (Partner//Sandbox)
 - Partner: SURF, ESnet and Géant
 - Graduate: HEAnet, NOMIOS, ShopVirge, GARR
 - Sandbox: Canarie, REANNZ, Internet2
- Join on discord →



Thank you!
so happy to be here :-)



Find me around for questions!